

GPU Baseline Results - Machine-Readable Figure Transcription

This document transcribes the values and labels visible in the supplied GPU-vs-FPGA baseline figure. Directly printed values are separated from values derived by multiplying a printed total by a printed percentage. Unlabeled stack-segment percentages are not guessed.

Legend / phase	Figure color
Transfer to GPU (P2P)	Blue
GPU preprocessing	Orange
GPU inference	Green
Publish + vFPGA polling	Yellow / amber

(a) End-to-end latency by phase

Bitstream size	End-to-end latency (ms)
64 KiB	0.19
256 KiB	0.20
1 MiB	0.25
4 MiB	0.45
16 MiB	1.24
64 MiB	4.39
128 MiB	8.59

Dashed reference line: **Paper, on-FPGA checker: 3.35 ms.**

(c) Compared with the paper's on-FPGA checker

Comparison item	Latency (ms)	What the source figure says
Paper: on-FPGA checker (U55C), preprocessing + inference	3.35	Printed horizontal-bar value
This work: GPU (MI210), preprocessing + inference	0.165	Printed horizontal-bar value
This work: end to end, 1 MiB bitstream	0.248	Printed horizontal-bar value

Experimental note printed below the figure: Medians. Transfer excludes card-memory (HBM) stalls. Panels (a), (b): size sweep, 200 trials per size; panel (c): 4 known-answer samples x 1000 trials.

Important transcription scope: The stacked bars contain additional colored segments whose numerical percentages are not individually printed. This PDF preserves the explicit labels and does not infer those missing percentages from bar geometry.

(b) Where the time goes

Size	Printed share label	Phase	Derived phase latency*
64 KiB	78%	GPU inference	0.148 ms
256 KiB	74%	GPU inference	0.148 ms
1 MiB	59%	GPU inference	0.147 ms
4 MiB	59%	Transfer to GPU (P2P)	0.266 ms
16 MiB	85%	Transfer to GPU (P2P)	1.054 ms
64 MiB	96%	Transfer to GPU (P2P)	4.214 ms
128 MiB	98%	Transfer to GPU (P2P)	8.418 ms

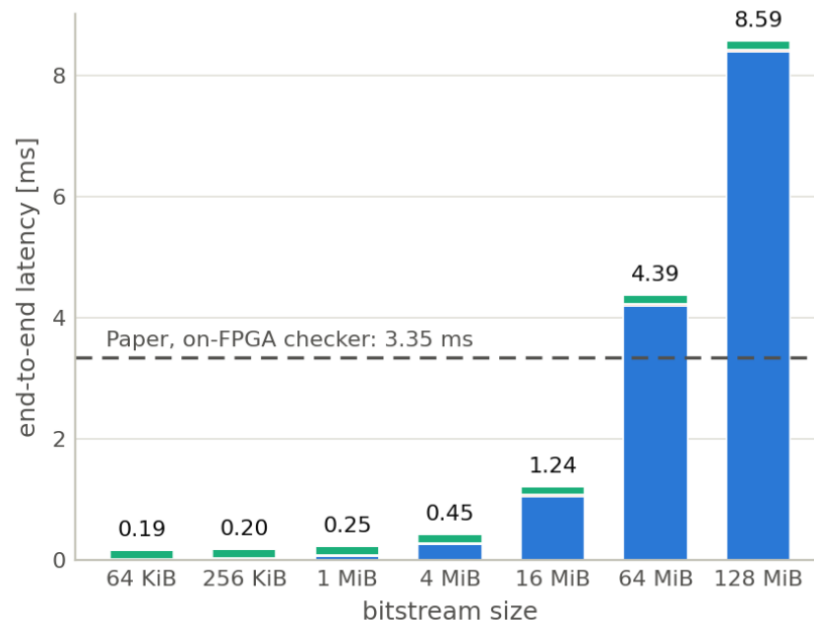
*Derived phase latency = panel (a) printed total x panel (b) printed percentage. These derived values are not separately printed in the source figure.

Source Figure

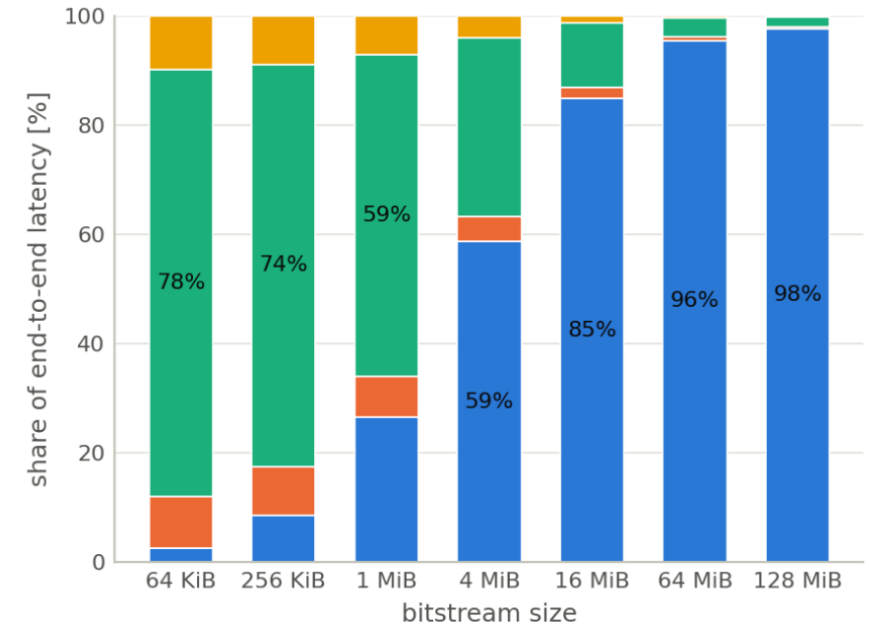
Original supplied figure, embedded for visual reference. The preceding page is the machine-readable transcription.

Transfer to GPU (P2P) GPU preprocessing GPU inference Publish + vFPGA polling

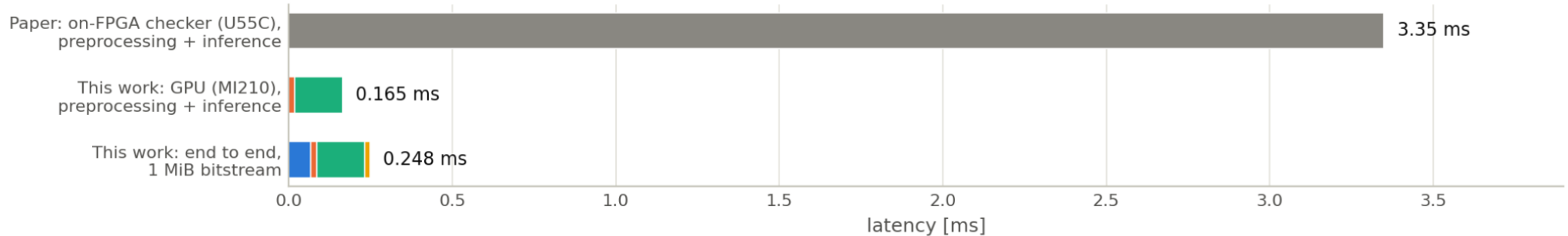
(a) End-to-end latency by phase



(b) Where the time goes



(c) Compared with the paper's on-FPGA checker



Medians. Transfer excludes card-memory (HBM) stalls. (a), (b): size sweep, 200 trials per size; (c): 4 known-answer samples $\times$ 1000 trials.